GG-P-125B
April 2, 1973
SUPERSEDING
Fed. Spec. GG-P-125A
May 4, 1965

FEDERAL SPECIFICATION

PEN, RULING, DRAFTSMAN'S: CONTOUR SWIVEL ACTION

This specification was approved by the Commissioner, Federal Supply Services, General Services Administration, for use of all Federal agencies.

1. SCOPE

1.1 This specification covers a swivel-action, spring-blade contour pen of the professional grade or quality.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications

QQ-C-586	- Copper-Nickel-Zinc Alloy; Rod, Shapes and Flat Products with Finished Edges (Flat Wire, Strip, and Bar).
QQ-S-766	- Steel Plates, Sheets, and Strip-Corrosion Resisting.
TT-I-528	- Ink, Drawing, Waterproof, Black.
UU-C-190	- Cardboard, Bristol, (Drafting).
UU-P-223	- Paper, Drawing.
UU-P-553	- Paper, Wrapping, Tissue.
CCC-C-531	- Cloth, Tracing.
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.

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PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-665	- Boxes; Paperboard, Metal Stayed (Including Stay Material).
PPP-B-676	- Boxes, Setup.
PPP-T-45	- Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing.

Federal Standards:

FED. STD. No. 123	- Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.

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(Copies of standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B122 - Copper-Nickel-Zinc Alloy (Nickel Silver) and Copper-Nickel Alloy Plate, Sheet, Strip, and Rolled Bar.
E18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street, NW, Washington, DC 20036.)

3. REQUIREMENTS

3.1 Description. The contour pen shall consist of a pen blade assembly attached to a thin rod which rotates in a tubular handle. The pen shall be constructed so that the blades are offset to the centerline of the handle and swivel without binding. A locking device shall hold the blades in a fixed position and a knurled screw shall permit adjustment of the blade points. The pen shall conform to the size and design shown in figure 1.

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3.2 Material. Material shall be as specified herein. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.2.1 Copper-nickel-zinc alloy. Copper-nickel-zinc alloy shall conform to ASTM B122 or QQ-C-586, composition 1, cold worked. Temper and form shall be optional.

3.2.2 Corrosion-resisting steel. Corrosion-resisting steel shall conform to QQ-S-766, class 301. Condition and temper shall be optional.

3.3 Blades. The pen blades shall be of the spring-blade type and shall be fabricated of corrosion-resisting steel or spring steel coated to prevent corrosion. The blade points shall have a hardness of Rockwell C55, plus or minus 5. The points shall be set, ground, and sharpened to permit drawing clean, unbroken, and uniform lines varying in width from 0.003 inch to 0.015 inch, without cutting into the drawing medium.

3.4 Rod. A corrosion-resisting steel rod shall be fastened securely to the pen blades (or made in one piece) to provide an axis for the swiveling action of the pen. The rod shall extend through the handle and shall be fitted so that swiveling action will be provided. The upper end of the rod shall be threaded and provided with a knurled nut of copper-nickel-zinc alloy or corrosion-resisting steel. Tightening of the nut shall clamp the rod and pen blades in a fixed, rigid position. When the rod is not clamped, the rod and blades shall swivel without binding and shall not have excessive play.

3.5 Handle. The tubular handle shall be of copper-nickel-zinc alloy, corrosion-resisting steel, aluminum, or brass with capped ends. Brass, if used, shall be either nickel-plated or black enameled. Aluminum, if used, shall be anodized.

3.6 Threads. All threads shall be of such size and pitch as to produce smooth action and shall withstand ordinary conditions of use.

3.7 Identification marking. Unless otherwise specified (see 6.2), each pen shall be identified in accordance with MIL-STD-130.

3.8 Finish. All exposed surfaces of the pen except the handle shall be highly polished or shall have a satin finish.

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3.9 Workmanship. Workmanship shall be of such grade or quality as to produce instruments of good balance, rigidity, and lightness equal to drafting instruments known commercially as the professional grade.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of preparation for delivery (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Sampling.

4.3.1.1 For examination. Sampling for examination shall be in accordance with MIL-STD-105.

4.3.1.2 For test. Sampling for tests shall be in accordance with MIL-STD-105, inspection level S-2.

4.3.2 Examination. Samples selected in accordance with 4.3.1.1 shall be examined as specified in 4.4.1. AQL shall be 4.0 percent defective.

4.3.3 Tests. Samples selected in accordance with 4.3.1.2 shall be tested as specified in 4.4.2. AQL shall be 1.5 percent defective.

4.4 Inspection procedure.

4.4.1 Examination. The pens shall be examined for the following defects:

101. Dimensions not as specified.
102. Material not as specified.

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103. Construction not as specified.
104. Lack of freedom in swiveling action.
105. Excessive play in swiveling action.
106. Blades cannot be clamped in fixed position.
107. Blades are not rigid when clamped in the fixed position.
108. Points cannot be adjusted to draw width of lines specified.
109. Workmanship not as specified.
110. Finish not as specified.
111. Identification marking not as specified.

4.4.2 Tests.

4.4.2.1 Hardness of points. The pen points shall be tested as specified in ASTM E18. Nonconformance to 3.3 shall constitute failure of this test.

4.4.2.2 Performance. The pen shall be filled with ink conforming to TT-I-528 and adjusted to produce a line 0.015 inch wide. A straight line, a sweeping or regular curved line, and a line containing not less than two "S" shaped figures or one "figure eight" shall be drawn free-hand on bristol board, drawing paper, or tracing cloth. The bristol board shall conform to UU-C-190, grade A, type II, class 1, ply optional. Drawing paper shall conform to UU-P-223, type I, class 1. Tracing cloth shall conform to CCC-C-531, type I. After the inked lines have dried, the lines shall be examined and measured. Nonconformance to 3.3 shall constitute failure of this test.

4.5 Inspection of preparation for delivery.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

112. Materials not as specified.
113. Unit packaging not as specified.
114. Intermediate quantities not as specified.
115. Strapping not zinc coated for level A.
116. Marking illegible, incomplete, or incorrect.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packaging. Each pen shall be wrapped with tissue paper conforming to UU-P-553, type II, class optional. Each wrapped pen shall be packaged in a box conforming to PPP-B-566, styles II, VI, or VII, variety, type and class optional or PPP-B-676, type, variety, class, and style optional.

5.1.1.2 Intermediate packaging. The packaged pens shall be intermediate packaged in quantities of 24 in boxes conforming to PPP-B-636, class weather-resistant, W5c. After closure, the boxes shall be waterproof sealed with tape in accordance with the appendix to the box specification.

5.1.2 Level B.

5.1.2.1 Unit packaging. The pens shall be unit packaged as specified in 5.1.1.1.

5.1.2.2 Intermediate packaging. The packaged pens shall be intermediate packaged in quantities of 24 in boxes conforming to PPP-B-566, all styles except XII and XIII, variety, type and class optional, PPP-B-665 class and style optional or PPP-B-676, type, variety, class and style optional. Closure of boxes shall be with tape conforming to PPP-T-45, type III, grade A.

5.1.3 Level C.

5.1.3.1 Unit packaging. Each pen shall be wrapped with neutral tissue paper. Each pen shall be packaged in a paperboard box or carton.

5.1.3.2 Intermediate packaging. The packaged pens shall be intermediate packaged in quantities of 24, in folding paperboard boxes or cartons.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. The pens, packaged as specified in 5.1, shall be packed in close-fitting boxes conforming to PPP-B-621, class 2, style optional; or PPP-B-601, overseas type, style optional. The gross weight of each box shall not exceed 200 pounds and the boxes shall be closed and strapped in accordance with the appendix to the applicable box specification. Strapping shall be zinc coated.

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5.2.2 Level B. The pens packaged as specified in 5.1 shall be packed in close-fitting boxes conforming to PPP-B-621, class 1, style optional, PPP-B-601, domestic type, style optional or PPP-B-636, V3s, V4s or V3c. The gross weight of each wood box shall not exceed 200 pounds and the gross weight of fiberboard boxes shall not exceed the weight limitation of the box specification. (For civil agencies, the packaged quantities shall be 240 pens and fiberboard boxes shall be class domestic.) Wood boxes shall be closed and strapped in accordance with the appendix to the applicable box specification. Weather resistant fiberboard boxes shall be waterproof sealed and reinforced with bands of reinforced tape in accordance with the appendix to the box specification. Domestic class fiberboard boxes shall be closed in accordance with method II of the appendix to PPP-B-636.

5.2.3 Level C. The pens, packaged as specified in 5.1, shall be packed to assure carrier acceptance and safe delivery to destination at lowest rates in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking. In addition to any special marking required by the contract or order, all marking shall be in accordance with 5.3.1 or 5.3.2.

5.3.1 Civil agencies. Packages and shipping containers shall be marked in accordance with FED. STD. No. 123.

5.3.2 Military agencies. Packages and shipping containers shall be marked in accordance with MIL-STD-129.

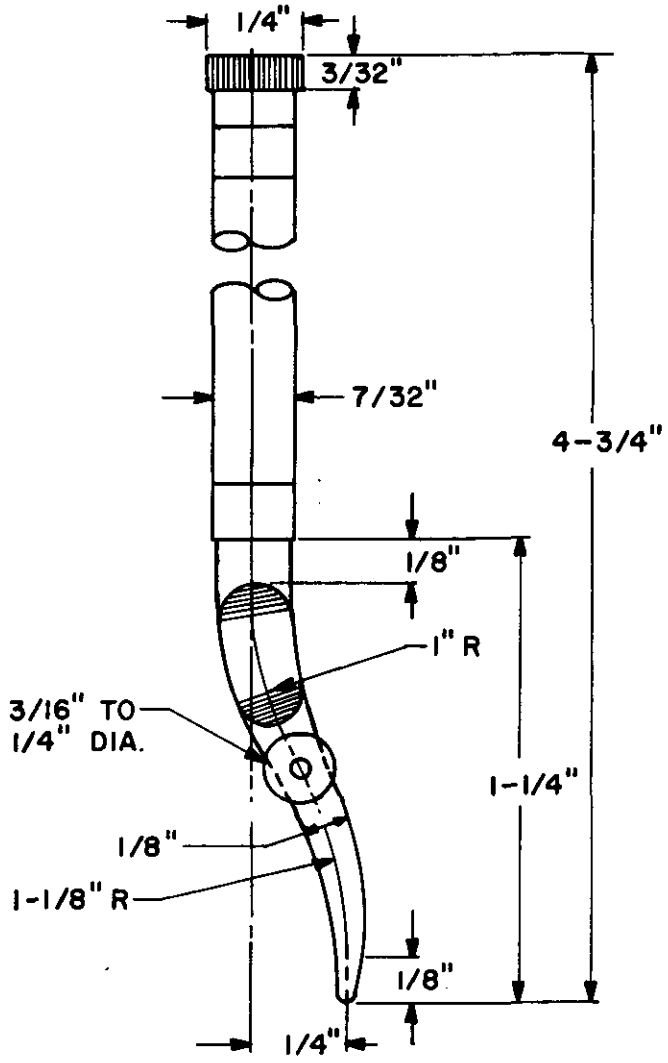
6. NOTES

6.1 Intended use. This pen is intended to be used for ruling irregular and regular ink lines and is especially designed for tracing contour lines on topographic maps.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Identification marking when other than as specified (see 3.7).
(c) Level of packaging and packing required (see 5.1 and 5.2). Level B packaging is intended to provide economical but limited protection and should be specified only when it is determined the pens will be held in covered storage for an indefinite period not to exceed 1 year from date of initial preservation and packaging.

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NOTE:
DECIMAL DIMENSIONS TO BE HELD WITHIN ± 0.01".
FRACTIONAL DIMENSIONS TO BE HELD WITHIN ± 1/32"
UNLESS OTHERWISE SPECIFIED.

Figure 1. CONTOUR PEN

X-2306

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MILITARY INTEREST:

Custodians:

Army - ME
Air Force - 82

User activities:

Navy - MC, YD

Preparing activity:

Army - ME

CIVIL AGENCY COORDINATING ACTIVITIES:

COMMERCE - NBS
GSA - FSS
JUSTICE - FPI

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION GG-P-125B Pen, Ruling, Draftsman's: Contour Swivel Action

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1426
11 Oct 72 (Rev)

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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